

Research article

Milk Physicochemical Properties of Holstein Friesian × Jersey Crossbred Cows under Different Feeding Management Systems in Tropical Southern Mindanao, Philippines

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Abstract

Effective feeding management is vital to improving milk quality in tropical dairy systems in the Philippines, where the domestic supply is limited. We evaluated how corn silage (CS), conventional fresh grass (FG), and mixed corn silage and spent grain (mixed CS & SG) feeding systems related to milk physicochemical properties in Holstein Friesian × Jersey crossbred cows on commercial farms in Southern Mindanao. Milk samples from 10 cows per farm were collected twice in May 2024 and analyzed for fat, SNF, density, lactose, protein, pH, and TSS. A two-way mixed ANOVA revealed farm-level effects ($\eta^2 = 0.568-0.799$) across all parameters. Mixed CS & SG feeding yielded superior SNF (8.48-8.66%), protein (3.09-3.15%), and density (29.65-30.37), while CS feeding produced the highest fat content (4.14-3.96%). FG feeding showed greater compositional variability. The collection period significantly affected lactose, pH, TSS, and milk temperature. Correlation analysis separated immediate temperature effects on density ($r = -0.610^*$) from chronic heat-stress responses, reducing milk solids (SNF: $r = -0.667^{**}$; protein: $r = -0.458^*$). Overall patterns reflected combined influences from feeding, management, and the environment rather than feeding alone. Nonetheless, silage-based systems supplemented with spent grain showed strong potential for improving milk quality in tropical smallholder dairy systems, with temperature-based insights guiding targeted management interventions.

Keywords: observational dairy study; feeding management systems; milk physicochemical properties; silage-spent grain supplementation; tropical dairy production; temperature stress effects

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1. Introduction

The Philippine dairy industry faces a supply–demand imbalance. Despite contributing 16.8% to total agricultural value and recent growth rates reaching 16.7% (PCAARRD, 2024), domestic milk production satisfies only about 1% of the country's estimated 2.5-billion-liter annual demand. This deficit necessitates reliance on imports, increasing trade vulnerability and economic exposure. National milk demand is projected to rise from one million to 1.8 million tons by 2029 (Department of Agriculture, 2024), underscoring the urgency of improving domestic production. However, production declines of 10.0% from April to June 2023 (Philippines Statistics Authority, 2023) reflect structural constraints, particularly among smallholder farmers who dominate and manage only 2–4 cattle (Bulatao, 2018).

Dairy production in Southeast Asia is constrained by environmental conditions that directly affect milk yield and quality. Elevated ambient temperatures (Thirumalai et al., 2017; Li, 2020), high relative humidity ($\geq 80\%$) (Kawakami & Hasebe, 2019; Thanh et al., 2023), and pronounced seasonal rainfall (Singh & Qin, 2020; Vaiglova et al., 2024) induce chronic heat stress, leading to reduced feed intake, altered metabolic activity, and impaired mammary synthesis (Vera et al., 2014; Hernández-Castellano et al., 2019; Coelho et al., 2020). These are compounded by feeding management challenges, seasonal variability in forage quality, limited availability of preserved feeds and supplements, inadequate storage, and insufficient knowledge of nutritionally balanced rations adapted to tropical conditions (Lagrange et al., 2015; Nyameasem et al., 2018). In Southern Mindanao, these challenges are intensified by Type III and IV climatic regimes, which constrain the ability to maintain consistent milk quality year-round.

Among management factors, feeding practices are the most adjustable determinant of milk physicochemical characteristics (Ganie et al., 2011; Sova et al., 2013). Nutrient supply governs the synthesis of milk fat, protein, lactose, and minerals through substrate availability and metabolic partitioning within the mammary gland (da Silva et al., 2017). Recent research has evaluated feeding strategies to improve milk quality under tropical conditions, including fresh forage–based cut-and-carry systems, silage-based feeding with preserved corn or sorghum to stabilize nutrient intake, and supplementation with agro-industrial by-products such as brewers' spent grain to enhance dietary protein and energy (Kung et al., 2018; Chetrariu & Dabija, 2023; Hayes et al., 2023). Each approach differs in nutrient consistency, cost, infrastructure, and suitability for smallholder systems. Evaluating how these feeding strategies translate into milk quality outcomes under commercial tropical conditions is critical for evidence-based extension and policy recommendations.

Despite extensive literature on the effects of feeding systems on milk quality, most evidence originates from temperate regions and intensive production systems, leaving a significant knowledge gap for smallholder dairying in the Philippines. Empirical data remain limited on the comparative performance of feeding systems under Southern Mindanao's specific climatic conditions, on milk physicochemical responses of Holstein Friesian $\times$ Jersey crossbreeds—the predominant genotype in the Philippines—and interaction between feeding management and thermal stress on milk composition. Moreover, few studies have assessed the effectiveness of feeding systems under Philippine smallholder farms.

Holstein Friesian $\times$ Jersey crossbreeds are adopted in tropical dairy systems due to their improved heat tolerance compared to pure Holsteins (Suadsong, 2012; Bereda et al., 2024), capacity to maintain milk production under elevated temperatures, balanced milk

composition, and suitability for smallholder operations (Bulatao, 2018; Habimana et al., 2023; Ndiokubwayo & Koç, 2023). However, characterization of milk quality in these crossbreeds under different feeding regimes remains scarce. Existing studies have centered on purebred temperate cattle or have not accounted for genotype-specific responses, limiting their relevance to Philippine production systems.

This observational comparative study examined associations between feeding management systems and milk physicochemical properties in Holstein Friesian × Jersey crossbreed cows under tropical conditions in Southern Mindanao, Philippines. It aims to compare milk quality, assess relationships between environmental temperature and milk physicochemical properties, and identify feeding practices with improved milk quality. By focusing on commercial farms rather than controlled experimental environments, this study provides insight relevant to smallholder dairy producers in Southeast Asia. It supports national efforts to enhance domestic milk quality, productivity, and self-sufficiency.

2. Materials and Methods

Before the study, we sought permits and complied with Republic Act No. 8485 (Animal Welfare Act of 1998), and the study was approved by the University of Southern Mindanao research ethics committee. This observational comparative study investigated milk physicochemical properties across three commercial dairy farms in Southern Mindanao, Philippines, practicing different feeding management systems: conventional fresh grass (FG), pure corn silage (CS), and combined corn silage with spent grain (mixed CS & SG). These systems had been practiced consistently for 5-8 years before sampling. The study was conducted in May 2024 across farms in Makilala, North Cotabato (Type III climate: shorter dry season November-April, temperatures 25-34°C, annual rainfall 2,000-2,800 mm, relative humidity 75-82%) and Polomolok and Tupi, South Cotabato (Type IV climate: evenly distributed rainfall, temperatures 27-32°C, annual rainfall 1,500-2,500 mm, relative humidity 78-85%).

We selected three commercial dairy farms in Southern Mindanao that kept Holstein Friesian × Jersey crossbreed cows. Selection was based on feeding systems, a minimum of 10 lactating cows (herd size), feeding practices that had been maintained for >3 years, and similar genetic composition across the farms. Ten lactating Holstein Friesian × Jersey crossbreed cows in each farm were sampled during both collection periods. At the same time, specific animal characteristics (age, parity, days in milk) were not controlled. Consistency was maintained by including only crossbred cows in active lactation, excluding cows with clinical mastitis or health issues, and sampling the same individual cows during both collection periods. Farm characteristics are summarized in Table 1.

On the Makilala farm (FG), cows were primarily fed fresh tropical grasses, predominantly Napier grass (*Pennisetum purpureum*), which was cut daily and offered twice daily with commercial concentrate at 30-35% of total dry matter intake plus mineral premixes in restricted feeding. The Polomolok farm (CS) provided corn silage as the primary forage, prepared on-farm by chopping whole-plant corn at appropriate moisture and ensiling in concrete bunker silos for at least 21 days, supplemented with commercial concentrate at 35-40% of total dry matter intake, offered twice daily in restricted amounts. On the Tupi farm, (mixed CS & SG) cows were fed corn silage and brewers' spent grain at a 50:50 ratio. Spent grain was sourced from local breweries and fed fresh within 2-3 days. The grain was supplemented with commercial concentrate at 30-35% of total dry matter intake, offered twice daily in restriction feeding. The commercial dairy concentrate was analyzed for nutritional composition at an accredited laboratory (Table 2). Primary forage

components were not subjected to laboratory analysis; however, nutritional profiles for these tropical forages in the Philippines were as follows: fresh Napier grass CP 10-13.83%, NDF 63-63.98%, ADF 35.60%, DM 14%; corn silage CP 6-7.06%, NDF 57-58.10%, DM 30-31.60%, ADF 31-33.34%; and brewers' spent grain CP 20-25%, NDF 40-50%, DM 20-25% (Kondo et al., 2015, Ysulat et al., 2024, Goyo et al., 2025; Lualhati et al., 2025).

Table 1. Characteristics of the three participating commercial dairy farms in Southern Mindanao, Philippines

Feeding Practices	Corn Silage (Polomolok)	Fresh Grass (Makilala)	Silage + Spent Grain (Tupi)
Location	South Cotabato	North Cotabato	South Cotabato
Climate type	Type IV	Type III	Type IV
Primary feeding system	Corn silage	Fresh grass	Corn silage + spent grain
Years practicing system	6 years	8 years	5 years
Housing system	Free-stall	Free-stall	Free-stall
Total milking cows	18	15	12
Cows sampled	10	10	10
Concentrate Supplementation	35-40% DM	30-35% DM	30-35% DM

Table 2. Nutritional composition of commercial dairy concentrate supplement used across all three farms

Nutrient Parameter	Composition (% or unit)
Crude Protein (CP)	16-18%
Dry Matter (DM)	88-90%
Neutral Detergent Fiber (NDF)	10-15%
Acid Detergent Fiber (ADF)	4-7%
Metabolizable Energy (ME)	12-12.8 MJ/kg DM
Supplementation Rate	30-40% of total DM
Mineral Premix	Included

Milk samples were collected twice at two-week intervals (May 5-7, 2024, and May 19-21, 2024). Approximately 100 mL composite milk samples were collected from each cow during morning milking sessions (0500-0600 h) immediately after milking into sterile pre-labeled bottles. Samples were immediately placed in ice-chilled cooler boxes (4-6°C) and transported to designated laboratory facilities purposely built within each farm. Physicochemical analysis was conducted within 2-3 h using a Lactoscan milk analyzer

calibrated according to manufacturer specifications, measuring fat content (%), solid nonfat (%), density, lactose (%), protein (%), pH, total soluble solids (%), and milk temperature (°C).

Environmental temperature and relative humidity were monitored at each farm using digital temperature-humidity meters (Model: HTC-1) stationed in barn/milking areas at cow-level height (approximately 1.5 m). Measurements were recorded twice daily: 2-3 hours before morning milking (0200-0300 h) and during milking (0500-0700 h). The reported values are the average of the two daily readings for each collection period. Milk temperature was automatically measured by the integrated sensor within the Lactoscan analyzer during sample analysis.

Statistical analysis was conducted using IBM SPSS Statistics with two-way mixed ANOVA for each milk parameter, treating dairy farms (three levels) as the between-subjects factor and collection period (two levels) as the within-subjects factor. Main effects, interactions, and effect sizes (partial η^2) were evaluated using Type III sums of squares with $\alpha = 0.05$. Post-hoc pairwise comparisons employed Bonferroni correction and Tukey's HSD test to identify specific farm differences. Pearson correlation coefficients were used to examine relationships between environmental temperature and milk physicochemical properties for each collection period.

3. Results and Discussion

3.1 Environmental conditions

Environmental temperature varied across the three study sites during both collection periods in May 2024. In the first collection, the highest temperature was recorded in Makilala at 23.06°C, followed by Polomolok at 22.28°C, while Tupi farm recorded the lowest at 21.89°C. Similar observation was recorded in the second collection, with Makilala posting 23.78°C, with a 0.72°C rise from the first collection. In the Polomolok site, the temperature rose to 23.08°C, 0.80°C higher than the first collection, while Tupi temperature slightly decreased to 21.32°C, down by 0.57°C.

3.2 Physicochemical analysis

3.2.1 Fats

Milk fat content was strongly affected by feeding, with a highly significant main effect of feeding practice ($P < 0.001$) and a very large effect size (partial $\eta^2 = 0.596$; Table 3). Corn silage (CS) feeding resulted in the highest milk fat content (4.05%), significantly exceeding FG (2.14%) and CS & SG (2.87%). Collection period also significantly influenced milk fat concentration ($P = 0.015$; partial $\eta^2 = 0.200$), with mean values declining from the first (3.17%) to the second collection (2.87%). The feeding practice $\times$ collection period interaction was not significant ($P = 0.150$; partial $\eta^2 = 0.131$).

The higher milk fat concentration under corn silage feeding reflects differences in rumen fermentation and substrate supply for mammary lipid synthesis. Milk fat synthesis relies on acetate and β -hydroxybutyrate derived from ruminal fermentation, with acetate contributing approximately 50% of the carbon for de novo fatty acid synthesis (Bauman & Griinari, 2003; Urrutia & Harvatine, 2017). Diets based on corn silage provide adequate neutral detergent fiber (NDF), promoting cellulolytic activity and acetate production, with acetate-to-propionate ratios of 2.5:1 to 3.5:1 favoring milk fat synthesis (Sutton et al., 2003;

Ferraretto & Shaver, 2015). In contrast, tropical fresh grasses may generate lower acetate-to-propionate ratios, particularly when high water-soluble carbohydrate contents accelerate propionate formation (Cabrita et al., 2003). Fermentation characteristics of tropical forages such as Napier grass (*Pennisetum purpureum*) are also highly variable and dependent on maturity stage (Surendra & Khanal, 2015; Li et al., 2023). The relatively stable composition of corn silage likely supported more consistent rumen fermentation and milk fat synthesis across collection periods, whereas fresh grass feeding is subject to diurnal and seasonal fluctuations that alter nutrient availability and fermentation dynamics (Bargo et al., 2003; Kung et al., 2018). The observed decline in milk fat across feeding systems (3.17% to 2.87%) further suggests the influence of heat stress. This effect may override the effects of dietary management alone in tropical environments (Bernabucci et al., 2010).

Table 3. Milk fat content of Holstein Friesian × Jersey cows under different feeding management at two sampling periods in Southern Mindanao

Feeding Practices	Collection 1	Collection 2	Overall by Feeding
CS-feeding	4.14±0.76 ^b	3.96±0.65	4.05±0.22 ^a
FG-feeding	2.45±1.23	1.83±0.63	2.14±0.22 ^b
Mixed CS & SG-feeding	2.91±0.46	2.82±0.54	2.87±0.22 ^b
Overall by collection	3.17±1.12 ^b	2.87±1.07 ^a	
P-Values & Effect sizes:			
Feeding system	P < 0.001** $\eta^2 = 0.596$		
Collection period	P = 0.015* $\eta^2 = 0.200$		
Feeding x Collection	P = 0.150 ^{ns} $\eta^2 = 0.131$		

CS – Corn silage; FG – Fresh grasses; CS & SG – Corn silage plus spent grain

Column and row mean and standard deviations by overall feeding and overall collection of the same letters are not statistically significant at the 0.05 level

** Highly significant P<0.001

* Significant P<0.05

^{ns} Not significant P>0.05

η^2 Effect size

3.2.2 Total soluble solids (TSS)

Milk total soluble solids (TSS) differed significantly among feeding management systems (P<0.001; partial $\eta^2 = 0.545$; Table 4). A significant interaction between feeding practice and collection period was also observed (p<0.001; partial $\eta^2 = 0.504$). Because the interaction was significant, the six feeding-by-period treatment combinations were compared using one-way ANOVA followed by Tukey's HSD post hoc test.

Among the six treatments, mixed CS & SG feeding in collection 2 produced the highest TSS (9.75 °Brix), significantly exceeding all other treatments except mixed CS & SG in collection 1 (9.15 °Brix; p = 0.096). FG-feeding in collection 2 recorded the lowest TSS (8.05 °Brix), significantly lower than all remaining treatments (p≤0.032). CS-feeding in both collection periods (8.75 and 8.85 °Brix) and FG-feeding in collection 1 (8.85 °Brix) were intermediate, with no significant differences detected among these three treatments (p≥0.483) (Table 4).

Table 4. Total soluble solids of milk from Holstein Friesian × Jersey cows under different feeding management at two sampling periods in Southern Mindanao

Feeding Practices		Collection 1		Collection 2		Overall by Feeding	
CS-feeding		8.75		8.85		8.80 ^b	
FG-feeding		8.85		8.05		8.45 ^a	
Mixed CS & SG-feeding		9.15		9.75		9.45 ^c	
<i>Feeding system</i>		p < 0.001**		η² = 0.545			
<i>Collection period</i>		p = 0.765 ^{ns}		η² = 0.003			
<i>Feeding × Collection</i>		p < 0.001**		η² = 0.504			
Six-treatment pairwise comparison (Tukey HSD)							
Treatment	CS × C1	CS × C2	FG × C1	FG × C2	Mix × C1	Mix × C2	Mean
CS × C1	—	-0.1	-0.1	+0.70*	-0.40	-1.00*	8.75 ^b
CS × C2	+0.10	—	0.00	+0.80*	-0.30	-0.90*	8.85 ^b
FG × C1	+0.10	0.00	—	+0.80*	-0.30	-0.90*	8.85 ^b
FG × C2	-0.70*	-0.80*	-0.80*	—	+1.10*	+1.70*	8.05 ^a
Mix × C1	+0.40	+0.30	+0.30	-1.10*	—	-0.60	9.15 ^b ^c
Mix × C2	+1.00*	+0.90*	+0.90*	-1.70*	+0.60	—	9.75 ^c

CS = corn silage; FG = fresh grass; Mix = mixed CS & SG; C1 = Collection 1; C2 = Collection 2. Values in pairwise cells are mean differences (I – J). * $p < 0.05$ (Tukey HSD). Cells with * marker indicate significant pairs.

The main effect of the collection period, averaged across feeding systems, was not significant ($p = 0.765$; partial $\eta^2 = 0.003$). Simple effects analysis revealed that feeding systems did not differ during collection 1 ($p = 0.192$) but diverged in collection 2 ($p < 0.001$; $\eta^2 = 0.678$). FG-feeding exhibited a significant decline in TSS between collection periods ($p < 0.001$), while mixed CS & SG-feeding showed a significant increase ($p = 0.004$). The numerical increase in TSS under CS-feeding between collection periods was non-significant (+0.100 °Brix; $p = 0.606$).

The superiority of the mixed CS & SG system reflects enhanced nutrient availability for milk solids synthesis. Brewers' spent grain is rich in rumen undegradable protein (RUP), with ruminal degradability of 42-52%, allowing dietary protein to bypass ruminal fermentation and be absorbed in the small intestine (Santos et al., 1984; Armentano et al., 1986; Westendorf & Wohlt, 2002). Direct intestinal amino acid absorption supports mammary gland protein synthesis without dependence on microbial protein production (Schingoethe, 1996). Inclusion of brewers' spent grain at 20-40% of dietary dry matter enhances feed efficiency without compromising milk protein concentration (Davis et al., 1983; Polan et al., 1985; West et al., 1994; Chitra, 2024). The fiber fraction of brewers'

spent grain, accounting for approximately 70% of its dry weight, provides fermentable substrate that sustains rumen function and volatile fatty acid production (White et al., 2008; Mitri et al., 2022; Lalowski et al., 2024), contributing to milk solids accumulation (Belibasakis & Tsirgogianni, 1996).

The significant feeding $\times$ collection interaction indicates that the advantages of the mixed CS & SG strategy and the vulnerability of exclusive FG-feeding became manifest only with continued dietary exposure. The absence of significant differences among feeding systems in collection 1 agrees with the progressive nature of ruminant dietary adaptation, whereby shifts in rumen microbial populations, fractional outflow rates, and mammary metabolism require sustained dietary input before producing measurable differences in milk composition (Benchaa et al., 2007; Dijkstra et al., 2011). Mixed CS & SG in collection 1 (9.15 °Brix) was not significantly different from either the intermediate group (CS-feeding and FG-feeding in collection 1; $p \geq 0.483$) or the highest treatment, mixed CS & SG in collection 2 (9.75 °Brix; $p = 0.096$), suggesting that the TSS benefit of supplying RUP from spent grain and fermentable energy from fresh grass was not yet fully expressed at initial sampling, becoming apparent only in collection 2 as cows adapted to the dietary treatment.

The decline in TSS under FG-feeding is consistent with the susceptibility of fresh-grass diets to temporal variation in milk solids output. Fresh-grass diets are characterized by variability in dry matter content, neutral detergent fiber concentration, water-soluble carbohydrate availability, and metabolizable energy density across seasonal and agronomic conditions (Fynn, 2012; Sautier et al., 2013; Pacheco et al., 2014; Fust & Schlecht, 2022). Fluctuations in forage quality between collection periods, compounded by inconsistencies in voluntary dry matter intake and animal body condition, likely reduced substrate availability for milk solids synthesis in the second collection. The temporal stability under CS-feeding agrees with observations that formulated concentrates attenuate the impact of environmental and management variation on milk composition (Davis et al., 1983; West et al., 1994). The interaction between feeding strategy and collection period, rather than feeding strategy alone, was the primary determinant of milk TSS in Holstein Friesian $\times$ Jersey cows under the conditions of this study. Mixed feeding incorporating locally available brewery byproducts offers a practical strategy for improving and stabilizing milk solids output across production cycles.

3.2.3 Milk density

Milk density was strongly influenced by feeding management ($P < 0.001$; partial $\eta^2 = 0.599$; Table 5), with feeding system accounting for 60% of the total variation in density. Mean density differed significantly among all feeding systems with the highest density observed under mixed CS & SG (30.02), followed by FG (27.75), while CS produced the lowest density (26.12). Neither the main effect of collection period ($P = 0.628$) nor the feeding practice $\times$ collection period interaction ($P = 0.693$) was significant.

The observed differences in milk density reflect variation of milk solids by lactose, protein, and minerals, with milk fat exerting a diluting effect due to its lower specific gravity (Walstra et al., 2006). The higher density under mixed CS & SG feeding aligns with its elevated total soluble solids. It suggests a greater accumulation of solids-not-fat components of protein and lactose, which may be supported by improved amino acid availability from bypass protein sources (Santos et al., 1984). In contrast, lower milk density under CS feeding, despite higher fat content (4.05%), suggests reduced non-fat solids, which was consistent with the inverse relationship between fat and milk density

(Samuelson et al., 2001). The intermediate density under FG feeding suggests partial compensation between lower fat concentration and moderate solids-not-fat content. The strong feeding effect and lack of temporal variation indicate that milk density is driven by feeding management rather than short-term environmental factors (Garcia et al., 2023). This stability makes milk density a reliable indicator for comparing milk quality across farms.

Table 5. Milk density of Holstein Friesian × Jersey cows under different feeding management at two sampling periods in Southern Mindanao

Feeding Practices	Collection 1	Collection 2	Overall by Feeding
CS-feeding	25.93±0.86	26.31±1.92	26.12±0.44 ^a
FG-feeding	27.92±3.06	27.58±2.17	27.75±0.44 ^b
Mixed CS & SG-feeding	29.66±1.58	30.38±1.48	30.02±0.44 ^c
Overall by collection	27.84±2.51	28.09±2.51	
P-Values & Effect sizes:			
Feeding system	P < 0.001** η^2 0.599		
Collection period	P = 0.628 ^{ns} η^2 0.009		
Feeding x Collection	P = 0.693 ^{ns} η^2 0.027		

CS – Corn silage; FG – Fresh grasses; CS & SG – Corn silage plus spent grain

Column mean and standard deviations by overall feeding of the same letters are not statistically significant at the 0.05 level

** Highly significant P<0.001

^{ns} Not significant P>0.05

η^2 Effect size

3.2.4 Percent lactose

Milk lactose content was significantly influenced by feeding management (P < 0.001; partial η^2 = 0.466; Table 6), with nearly half of the observed variation attributable to differences among feeding systems. Cows fed mixed CS & SG produced milk with a higher lactose concentration (4.71%) than those receiving CS (4.28%) or FG (4.33%). Neither the main effect of collection period (P = 0.402) nor the feeding practice × collection period interaction (P = 0.923) was significant.

The elevated lactose concentration under mixed CS & SG feeding reflects improved energy balance and glucose availability for mammary lactose synthesis. Lactose is synthesized in the mammary gland from blood glucose, and is maintained within a narrow range across nutritional conditions (Fox & McSweeney, 1998). In ruminants, glucose for lactose synthesis is derived from hepatic gluconeogenesis, with propionate contributing about 50-60% of total glucose production (Reynolds et al., 2003). Fermentation of residual starch and fermentable fiber in brewers' spent grain may enhance propionate production, thereby increasing glucose availability for lactose synthesis (Westendorf & Wohlt, 2002). In addition, the bypass protein supplied by spent grain reduces the energetic cost of milk protein synthesis by providing preformed amino acids, allowing greater glucose partitioning toward lactose production (Schingoethe, 1996). The absence of temporal effects reflects

the physiological regulation of lactose as the primary osmotic determinant of milk volume, with concentration maintained to stabilize milk secretion and osmotic pressure (Linzell & Peaker, 1971). The significant differentiation observed—mixed CS & SG versus CS and FG—contrasts with the differentiation patterns reported for fat, TSS, and density, indicating that lactose responds to nutritional enhancement in a threshold-dependent rather than dose-dependent manner.

Table 6. Milk lactose of Holstein Friesian × Jersey cows under different feeding management at two sampling periods in Southern Mindanao

Feeding Practices	Collection 1	Collection 2	Overall by Feeding
CS-feeding	4.25±0.11	4.30±0.29	4.28±0.06 ^a
FG-feeding	4.32±0.41	4.346±0.35	4.33±0.06 ^a
Mixed CS & SG-feeding	4.66±0.21	4.76±0.23	4.71±0.06 ^b
Overall by collection	4.41±0.32	4.47±0.35	
P-Values & Effect sizes:			
Feeding system	P < 0.001** η^2 0.466		
Collection period	P = 0.402 ^{ns} η^2 0.026		
Feeding x Collection	P = 0.923 ^{ns} η^2 0.006		

CS – Corn silage; FG – Fresh grasses; CS & SG – Corn silage plus spent grain

Column mean and standard deviations by overall feeding of the same letters are not statistically significant at the 0.05 level

** Highly significant P<0.001

^{ns} Not significant P>0.05

η^2 Effect size

3.2.5 Milk protein

Milk protein content differed significantly among feeding systems (P<0.001; partial η^2 = 0.463; Table 7), with nearly half of the total variation attributable to feeding differences. Cows fed mixed CS & SG produced milk with higher protein content (3.13%) than those fed FG (2.84%) or CS (2.88%). Neither the main effect of collection period (P = 0.540) nor the feeding practice × collection period interaction (P = 0.873) was significant.

Brewers' spent grain (BSG) has relatively low rumen degradability (42-52%), allowing a greater proportion of dietary protein to bypass rumen breakdown and be absorbed in the intestine as amino acids for milk protein synthesis (Chalupa, 1975; Promkot & Wanapat, 2009; Singh et al., 2019). This differs from highly degradable protein sources such as soybean meal (65-75%), which contribute more to microbial protein synthesis (Aufrere et al., 1994; Maxin et al., 2013). BSG is enriched in insoluble barley storage proteins following brewing and shows moderate digestibility (71-88%) in animal diets (Sauvant et al., 2004; Jaeger et al., 2021; Devnani et al., 2023 ; Nyhan et al., 2023). Studies consistently report maintained or increased milk protein concentration and improved feed efficiency with BSG inclusion (Johnson et al., 1987; West et al., 1994; Belibasakis & Tsirgogianni, 1996; Mahnken, 2010; Chitra, 2024). The lack of temporal variation in milk protein suggests physiological buffering against short-term dietary

changes (Attaix et al., 2005; Baracos, 2006; Bröer & Bröer, 2017; McCabe & Boerman, 2020 ; Tomé et al., 2020). The response pattern indicates that the metabolizable protein supply in the mixed CS & SG system met the requirements for maximal milk protein synthesis, whereas it remained below this level in the CS and FG systems.

Table 7. Milk protein of Holstein Friesian × Jersey cows under different feeding management at two sampling periods in Southern Mindanao

Feeding Practices	Collection 1	Collection 2	Overall by Feeding
CS-feeding	2.84±0.06	2.846±0.20	2.84±0.45 ^a
FG-feeding	2.87±0.28	2.884±0.24	2.88±0.45 ^a
Mixed CS & SG-feeding	3.10±0.14	3.157±0.15	3.13±0.45 ^b
Overall by collection	2.93±0.21	2.96±0.24	
P-Values & Effect sizes:			
Feeding system	P < 0.001** η^2 0.463		
Collection period	P = 0.540 ^{ns} η^2 0.014		
Feeding x Collection	P = 0.873 ^{ns} η^2 0.010		

CS – Corn silage; FG – Fresh grasses; CS & SG – Corn silage plus spent grain

Column mean and standard deviations by overall feeding of the same letters are not statistically significant at the 0.05 level

** Highly significant P<0.001

^{ns} Not significant P>0.05

η^2 Effect size

3.2.6 pH of milk

Milk pH was not significantly affected by feeding system ($p = 0.276$; partial $\eta^2 = 0.091$), collection period ($p = 0.167$; partial $\eta^2 = 0.070$), or the feeding system × collection period interaction ($p = 0.048$; partial $\eta^2 = 0.202$; Table 8). Although the interaction term reached significance, the effect size was small, and comparison of the six feeding-by-period treatment combinations using one-way ANOVA and Tukey's HSD post hoc test revealed no significant differences among any of the 15 pairs (all pairwise $p \geq 0.073$). All six treatment means were on a single homogeneous subset, indicating that no treatment combination produced a milk pH distinguishable from that of any other treatment combination. Cell means ranged from 6.832 (FG-feeding × Collection 2) to 6.923 (mixed CS & SG × Collection 1), a spread of 0.091 pH units across all treatments and periods (Figure 1). Between collections, pH increased numerically under CS-feeding (+0.047) but declined under FG-feeding (−0.068) and mixed CS & SG-feeding (−0.067); none of these within-group changes were significant after Bonferroni correction.

Milk pH reflects the balance between acid–base buffering systems in milk, primarily caseins, phosphate salts, citrate, and bicarbonate, with regular bovine milk typically exhibiting pH values between 6.5 and 6.8 (Tsioulpas et al., 2007a,b). The elevated pH range observed in this study (6.80–7.00) may be attributed to measurement and sample-handling errors or to breed-specific characteristics of Holstein Friesian × Jersey crossbred cows. Unlike milk fat, protein, lactose, and density, milk pH was more responsive to short-term, temporal influences, as reflected by the significant interaction effect. Milk pH is known

to be influenced by the stage of lactation (Sapru et al., 1997; Coulon et al., 1998; Abduljalil & Faisal, 2024), udder health status (Sloth et al., 2003; Shinozuka et al., 2018), milking interval (Fernando & Spahr, 1983; Kelly et al., 1998), and the metabolic state of the cow before milking (Henno et al., 2008). The absence of significant pairwise differences among the six treatment combinations, despite a significant interaction term, suggests that pH variation across feeding systems and collection periods was insufficient to exceed the family-wise error threshold of Tukey's HSD correction, and is unlikely to carry biological relevance given the narrow range of pH values recorded across treatments. The convergence of pH values during the second collection, regardless of feeding system, suggests that cows adapted to prevailing thermal and management conditions between sampling periods, or that transient, farm-specific influences present during collection 1 were resolved by collection 2. Increased within-group variability, particularly under mixed CS & SG-feeding in collection 2, may reflect differences in milk handling, storage duration, or microbial activity if samples were not preserved under consistent temperature and storage conditions. Milk pH was not responsive to feeding management under the conditions of this study, and standardization of sampling, handling, and analytical procedures remains essential to minimize residual variation in observational field studies where environmental and management conditions cannot be fully controlled.

Table 8. pH of milk of Holstein Friesian × Jersey cows under different feeding management at two sampling periods in Southern Mindanao

Feeding practices	Collection 1	Collection 2	Overall by feeding				
CS-feeding	6.88	6.92	6.90				
FG-feeding	6.90	6.83	6.87				
Mixed CS & SG-feeding	6.92	6.86	6.90				
Feeding system	< 0.276 ^{ns} η^2 0.091						
Collection period	0.167 ^{ns} η^2 0.070						
Feeding $\times$ Collection	< 0.048 ^{**} η^2 0.202						
Six-treatment pairwise comparison (Tukey HSD)							
Treatment	CS $\times$ C1	CS $\times$ C2	FG $\times$ C1	FG $\times$ C2	Mix $\times$ C1	Mix $\times$ C2	Mean
CS $\times$ C1	—	-0.047	-0.025	0.043	-0.048	0.019	6.87
CS $\times$ C2	0.066	—	0.047	0.022	0.09	-0.001	6.92
FG $\times$ C1	-0.023	0.044	—	0.025	-0.022	0.068	6.90
FG $\times$ C2	-0.068	-0.091	-0.024	—	-0.043	-0.09	6.83
Mix $\times$ C1	0.001	0.023	0.091	0.067	—	0.048	6.92
Mix $\times$ C2	-0.019	-0.066	-0.044	0.024	-0.067	—	6.86

CS = corn silage ; FG = fresh grass ; Mix = mixed CS & SG ; C1 = Collection 1 ; C2 = Collection 2. Values in pairwise cells are mean differences (I – J).

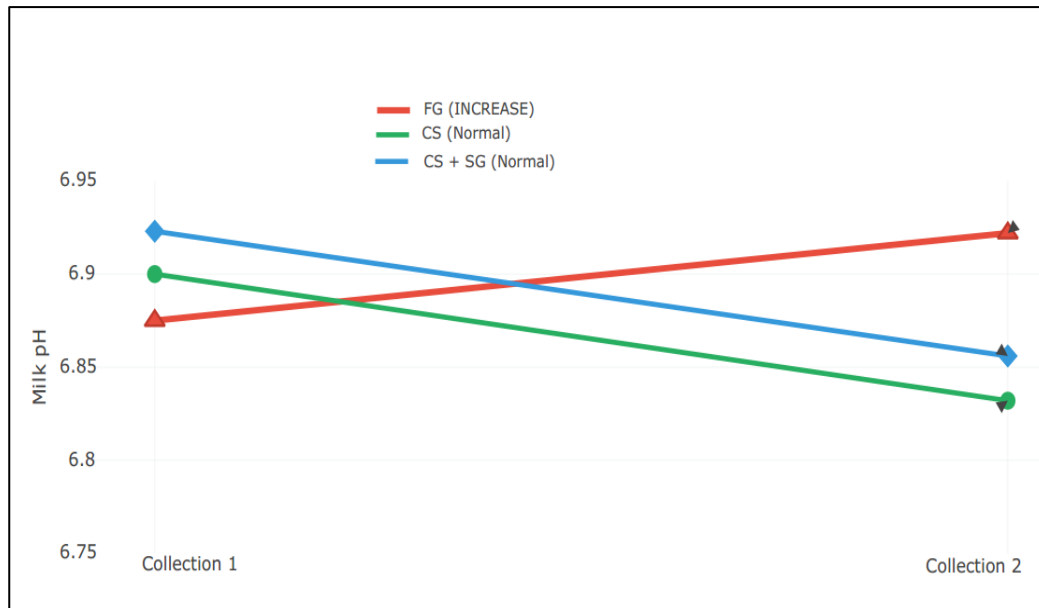


Figure 1. Interaction between dairy farm feeding practices and collection period on pH of milk

3.2.7 SNF

Milk solids-not-fat (SNF) ranged from 7.80% to 8.66% across farms and sampling periods. Feeding management had a strong effect on SNF ($P < 0.001$; partial $\eta^2 = 0.514$; Table 9). Milk from cows fed corn silage plus spent grain (Mixed CS & SG) had higher SNF (8.57%) than that from corn silage (7.81%) or fresh grass (7.98%), which did not differ. The collection period and its interaction with the feeding system were not significant.

Solids-not-fat represents the combined milk proteins, lactose, minerals, and vitamins and therefore serves as an indicator of milk nutritional density and processing value (Walstra et al., 2006). The higher SNF concentration observed under Mixed CS & SG feeding reflects the cumulative enhancement of milk protein and lactose synthesis driven by improved metabolizable protein and energy supply. The clear difference between mixed CS & SG and the CS and FG systems corroborate the patterns observed for protein and lactose, confirming SNF as an integrated indicator of milk solids. Higher SNF under the mixed CS & SG feeding is consistent with the protein contribution of brewers' spent grain, which contains 25-34% crude protein primarily in rumen-undegradable form (Westendorf & Wohlt, 2002), and has been shown to increase total milk solids under hot conditions (Belibasakis & Tsirgogianni, 1996). The lack of temporal variation indicates that SNF is relatively stable and buffered against short-term environmental changes. Together, the significant effect size and temporal stability support SNF as a reliable indicator of long-term nutritional management, with the consistent advantage of mixed CS & SG feeding highlighting the value of bypass protein supplementation in tropical dairy systems.

Table 9. Milk SNF of Holstein Friesian × Jersey cows under different feeding management at two sampling periods in Southern Mindanao

Feeding Practices	Collection 1	Collection 2	Overall by Feeding
CS-feeding	7.80±0.17	7.83±0.53	7.81±0.11 ^a
FG-feeding	8.06±0.31	7.91±0.65	7.98±0.11 ^a
Mixed CS & SG-feeding	8.49±0.38	8.66±0.42	8.57±0.11 ^b
Overall by collection	8.11±0.41	8.13±0.65	
P-Values:			
Feeding system	P < 0.001** η^2 0.514		
Collection period	P = 0.873 ^{ns} η^2 0.001		
Feeding x Collection	P = 0.446 ^{ns} η^2 0.058		

CS – Corn silage; FG – Fresh grasses; CS & SG – Corn silage plus spent grain
 Column mean and standard deviations by overall feeding of the same letters are not statistically significant at the 0.05 level

** Highly significant P<0.001

^{ns} Not significant P>0.05

η^2 Effect size

3.2.8 Milk temperature

Milk temperature varied widely across farms (14.6–30.6°C) and was strongly influenced by farm factors (P<0.001; partial η^2 = 0.799; Table 10). Mean temperatures were highest under fresh grass feeding (29.18°C), intermediate under corn silage (18.24°C), and lowest under corn silage plus spent grain (14.80°C). Milk from fresh grass systems was significantly warmer than that from the other systems. Collection period effects were not significant, and relative differences among farms remained consistent over time.

Milk temperature is an indicator of post-milking handling efficiency and cold-chain integrity, with direct implications for microbial growth, shelf life, and overall milk quality. International standards recommend cooling milk to ≤4°C within 2-3 h after milking to suppress bacterial proliferation (Poli, 2004). The temperatures observed in this study exceeded these thresholds, reflecting common limitations in post-harvest cooling within tropical smallholder dairy systems. The consistently high temperatures under FG feeding suggest minimal or delayed cooling, creating conditions favorable for rapid microbial multiplication, with bacterial populations doubling every 20-30 min at temperatures above 25°C (Elmoslemany et al., 2009). Although CS systems exhibited lower temperatures than FG, values remained above standards, indicating insufficient cooling capacity. Mixed CS & SG systems maintained lower temperatures, suggesting better cooling infrastructure or handling practices, though still below industry requirements. Elevated milk temperatures accelerate microbial growth and enzymatic activity, particularly of lipases and proteases, leading to physicochemical deterioration and reduced processing suitability (Sørhaug & Stepaniak, 1997). The dominance of farm-level effects underscores the potential for improvements in cooling infrastructure, standardized temperature monitoring, and improved milk-handling, which are essential for translating compositional gains into consistent quality outcomes across the dairy value chain.

Table 10. Milk temperature of Holstein Friesian × Jersey cows under different feeding management at two sampling periods in Southern Mindanao

Feeding Practices	Collection 1	Collection 2	Overall by Feeding
CS-feeding	17.24±1.93	19.23±4.93	18.24±1.02 ^a
FG-feeding	27.72±4.91	30.63±5.56	29.18±1.04 ^b
Mixed CS & SG-feeding	14.99±2.53	14.61±1.49	14.80±1.04 ^a
Overall by collection	19.98±6.62	21.49±8.04	
P-Values:			
Feeding system	P < 0.001** η^2 0.799		
Collection period	P = 0.069 ^{ns} η^2 0.117		
Feeding x Collection	P = 0.239 ^{ns} η^2 0.100		

CS – Corn silage; FG – Fresh grasses; CS & SG – Corn silage plus spent grain
 Column mean and standard deviations by overall feeding of the same letters are not statistically significant at the 0.05 level

** Highly significant P<0.001

^{ns} Not significant P>0.05

η^2 Effect size

3.2.9 Correlation analysis

Correlation analysis distinguished two temperature-related effects on milk quality: immediate physical responses and longer-term physiological adaptations (Table 11). Milk temperature showed strong negative correlations with density across both collection periods ($r = -0.610^*$), reflecting thermal expansion effects whereby liquid density decreases as temperature increases (McCarthy & Singh, 2009; Suhendra et al., 2020).

In contrast, negative associations between milk temperature and compositional traits such as solids-not-fat (SNF; $r = -0.667^*$) and protein ($r = -0.458^*$) indicate chronic physiological responses to sustained heat exposure. These patterns are consistent with heat stress–related reductions in feed intake, altered nutrient partitioning, and metabolic adjustments that limit substrate availability for milk synthesis (Wheelock et al., 2010; Kumar et al., 2020; Sammad et al., 2020).

Several traits showed evidence of temporal adaptation. Positive correlations between temperature and fat weakened over time, while associations involving lactose and pH similarly diminished, suggesting partial acclimatization to thermal conditions (Bernabucci et al., 2010). In contrast, total soluble solids (TSS) showed an emerging negative relationship with temperature, indicating cumulative effects of prolonged heat exposure.

Table 11. Correlation analysis between environmental temperature and physicochemical values during first and second collection period in Southern Mindanao, Philippines

Parameter	1st Collection	2nd Collection	Interpretation
% FAT	0.538**a	0.265 ^{ns}	Positive; weakened over time
% SNF	-0.667**b	-0.580**b	Consistently negative
Density	-0.610**b	-0.689**b	Consistently negative; strengthened over time
% Lactose	-0.484**b	0.007 ^{ns}	Negative; weakened over time
% Protein	-0.458* ^b	-0.584**b	Consistently negative; strengthened over time
pH	-0.493**b	0.193 ^{ns}	Negative; weakened over time
TSS	-0.302 ^{ns}	-0.591**b	Negative; emerged in 2 nd collection
Milk Temperature	-0.020 ^{ns}	0.429* ^a	Absent in 1st collection; positive in 2 nd collection

** p<0.01; * p<0.05; ^{ns} not significant (p≥0.05). ^a Positive correlation: both variables change in the same direction. ^b Negative correlation: variables change in opposite directions. All correlations are between environmental temperature and the listed milk quality parameter.

These findings have practical implications for management. Physical temperature effects, such as those on density, can be mitigated through improved post-milking cooling. In contrast, compositional declines require animal-level strategies, including nutritional optimization, heat abatement, and feeding systems such as corn silage supplemented with spent grain. Together, the results highlight the need for targeted interventions to maintain milk quality under tropical conditions.

4. Conclusions

This observational study demonstrated that feeding management represents the primary modifiable factor influencing milk composition in Holstein Friesian × Jersey crossbreeds under tropical conditions, with farm-level effects explaining 46-80% of variation across parameters. Corn silage feeding consistently produced superior milk fat content (4.05%), while the combined silage and spent grain system yielded highest protein (3.13%), lactose (4.71%), solids-not-fat (8.57%), and density (30.02), reflecting enhanced bypass protein supply and nutrient density. Correlation analyses distinguished immediate physical temperature effects on density from chronic physiological responses affecting milk solids, indicating that effective tropical dairy management requires both rapid cooling infrastructure and integrated nutritional strategies. Fresh grass systems exhibited greater compositional variability and lower solids content compared with silage-based approaches, suggesting that preserved forages with protein supplementation offer enhanced stability for smallholder tropical dairy operations. These findings provide preliminary evidence supporting silage-based feeding strategies for improving milk quality in resource-constrained tropical environments.

5. Authors' Contributions

Author R.I.Y. contributes to conceptualization and data curation. Author A.N.T. contributes to the formal analysis and writing the original draft. Author E.C.T. contributes to investigation/conduct, review, and editing.

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6. Conflicts of Interest

The authors have not declared any conflict of interest.

References

- Abduljalil, K., & Faisal, M. (2024). Impact of milk pH and fat content on the prediction of milk-to-plasma ratio: Knowledge gap and considerations for lactation study design and interpretation. *Clinical Pharmacokinetics*, 63(11), 1561-1572. <https://doi.org/10.1007/s40262-024-01432-w>
- Armentano, L. E., Herrington, T. A., Polan, C. E., Moe, A. J., Herbein, J. H., & Umstadt, P. (1986). Ruminal degradation of dried brewers grains, wet brewers grains, and soybean meal. *Journal of Dairy Science*, 69(8), 2124-2133. [https://doi.org/10.3168/jds.S0022-0302\(86\)80644-0](https://doi.org/10.3168/jds.S0022-0302(86)80644-0)
- Attaix, D., Mosoni, L., Dardevet, D., Combaret, L., Mirand, P. P., & Grizard, J. (2005). Altered responses in skeletal muscle protein turnover during aging in anabolic and catabolic periods. *The International Journal of Biochemistry and Cell Biology*, 37(10), 1962-1973. <https://doi.org/10.1016/j.biocel.2005.04.009>
- Aufrere, J., Graviou, D., & Michalet-Doreau, B. (1994). Degradation in the rumen of proteins of 2 legumes: soybean meal and field pea. *Reproduction Nutrition Development*, 34(5), 483-490. <https://doi.org/10.1051/rnd:19940509>
- Baracos V. E. (2006). Integration of amino acid metabolism during intense lactation. *Current Opinion in Clinical Nutrition and Metabolic Care*, 9(1), 48-52. <https://doi.org/10.1097/01.mco.0000198059.72872.c8>
- Bargo, F., Muller, L. D., Kolver, E. S., & Delahoy, J. E. (2003). Invited review: Production and digestion of supplemented dairy cows on pasture. *Journal of Dairy Science*, 86(1), 1-42. [https://doi.org/10.3168/jds.s0022-0302\(03\)73581-4](https://doi.org/10.3168/jds.s0022-0302(03)73581-4)
- Bauman, D. E., & Griinari, J. M. (2003). Nutritional regulation of milk fat synthesis. *Annual Review of Nutrition*, 23(1), 203-227.
- Belibasakis, N. G., & Tsirgogianni, D. (1996). Effects of wet brewers grains on milk yield, milk composition and blood components of dairy cows in hot weather. *Animal Feed Science and Technology*, 57(3), 175-181.
- Bereda, A., Agune, B., & Yilma, Z. (2024). Performance and farmers preferred traits of Jersey × local cows in the tropics: the case of *Angolelana Tera* district, North Shewa zone, Ethiopia. *Cogent Social Sciences*, 10(1), Article 2368948. <https://doi.org/10.1080/23311886.2024.2368948>
- Benchaa, C., Chaves, A. V., Fraser, G. R., Beauchemin, K. A., & McAllister, T. A. (2007). Effects of essential oils and their components on in vitro rumen microbial fermentation. *Canadian Journal of Animal Science*, 87(3), 413-419. <https://doi.org/10.4141/CJAS07012>
- Bernabucci, U., Lacetera, N., Baumgard, L. H., Rhoads, R. P., Ronchi, B., & Nardone, A. (2010). Metabolic and hormonal acclimation to heat stress in domesticated ruminants. *Animal*, 4(7), 1167-1183. <https://doi.org/10.1017/S175173111000090X>

- Bröer, S., & Bröer, A. (2017). Amino acid homeostasis and signalling in mammalian cells and organisms. *Biochemical Journal*, 47(4), 1935-1963. <https://doi.org/10.1042/BCJ20160822>
- Bulatao, S. M. (2018). *Philippines: Promoting dairy entrepreneurship through enterprise zones*. <https://www.fao.org/4/i0588e/i0588e09.htm>
- Cabrita, A. R. J., Fonseca, A. J. M., Dewhurst, R. J., & Gomes, E. (2003). Nitrogen supplementation of corn silages. 2. Assessing rumen function using fatty acid profiles of bovine milk. *Journal of Dairy Science*, 86(12), 4020-4032.
- Chalupa, W. (1975). Rumen bypass and protection of proteins and amino acids. *Journal of Dairy Science*, 58(8), 1198-1218. [https://doi.org/10.3168/jds.S0022-0302\(75\)84697-2](https://doi.org/10.3168/jds.S0022-0302(75)84697-2)
- Chetrariu, A., & Dabija, A. (2023). Spent grain: A functional ingredient for food applications. *Foods*, 12(7), Article 1533. <https://doi.org/10.3390/foods12071533>
- Chitra, P. (2024). Effect of dietary supplementation of wet brewer's grains on lactation performance of Jersey cross breed cows. *International Journal of Agriculture, Environment and Biotechnology*, 17(2), 155-160. <https://doi.org/10.30954/0974-1712.02.2024.9>
- Coelho, M. G., Silva, F. L., Silva, M. D., Silva, A. P., Cezar, A. M., Slanzon, G. S., Miqueo, E., Toledo, A. D., & Bittar, C. M. (2020). Acidified milk for feeding dairy calves in tropical raising systems. *Journal of Animal and Feed Sciences*, 29, 215-223. <https://www.doi.org/10.22358/jafs/127690/2020>
- Coulon, J.-B., Verdier, I., Pradel, P., & Almena, M. (1998). Effect of lactation stage on the cheesemaking properties of milk and the quality of Saint-Nectaire-type cheese. *The Journal of Dairy Research*, 65(2), 295-305. <https://doi.org/10.1017/s002202999700277x>
- da Silva, T. C., da Silva, L. D., Santos, E. M., Oliveira, J. S., & Perazzo, A. F. (2017). Importance of the fermentation to produce high-quality silage. *Fermentation Processes*, 8, 1-20. <https://doi.org/10.5772/64887>
- Davis, C.L., Grenawalt, D., & McCoy, G.C. (1983). Feeding value of pressed brewers' grains for lactating dairy cows. *Journal of Dairy Science*, 66, 73-79. [https://doi.org/10.3168/JDS.S0022-0302\(83\)81755-X](https://doi.org/10.3168/JDS.S0022-0302(83)81755-X)
- Department of Agriculture (2024, October). *NDA unveils comprehensive plan for Philippine dairy industry growth*. <https://www.da.gov.ph/nda-unveils-comprehensive-plan-for-philippine-dairy-industry-growth/>
- Devnani, B., Moran, G. C., & Grossmann, L. (2023). Extraction, composition, functionality, and utilization of brewer's spent grain protein in food formulations. *Foods*, 12(7), Article 1543. <https://doi.org/10.3390/foods12071543>
- Dijkstra, J., Van Zijderveld, S., Apajalahti, J., Bannink, A., Gerrits, W., Newbold, J., Perdok, H., & Berends, H. (2011). Relationships between methane production and milk fatty acid profiles in dairy cattle. *Animal Feed Science and Technology*, 166-167, 590-595. <https://doi.org/10.1016/j.anifeedsci.2011.04.042>
- Elmoslemany, A., Keefe, G., Dohoo, I., & Jayarao, B. (2009). Risk factors for bacteriological quality of bulk tank milk in Prince Edward Island dairy herds. Part 1: Overall risk factors. *Journal of Dairy Science*, 92(6), 2634-2643. <https://doi.org/10.3168/jds.2008-1812>
- Fernando, R. S., & Spahr, S. L. (1983). Effects of milking interval on selected milk constituents from normal and infected quarters. *Journal of Dairy Science*, 66(5), 1155-1161. [https://doi.org/10.3168/jds.S0022-0302\(83\)81912-2](https://doi.org/10.3168/jds.S0022-0302(83)81912-2)
- Ferraretto, L. F., & Shaver, R. D. (2015). Effects of whole-plant corn silage hybrid type on intake, digestion, ruminal fermentation, and lactation performance by dairy cows through a meta-analysis. *Journal of Dairy Science*, 98(4), 2662-2675.
- Fox, P. F., Uniacke-Lowe, T., McSweeney, P. L. H., & O'Mahony, J. A. (2015). *Dairy chemistry and biochemistry* (2nd ed.). Springer. <https://doi.org/10.1007/978-3-319-14892-2>

- Fust, P., & Schlecht, E. (2022). Importance of timing: Vulnerability of semi-arid rangeland systems to increased variability in temporal distribution of rainfall events as predicted by future climate change. *Ecological Modelling*, 468, Article 109961. <https://doi.org/10.1016/j.ecolmodel.2022.109961>
- Fynn, R. W. (2012). Functional resource heterogeneity increases livestock and rangeland productivity. *Rangeland Ecology and Management*, 65(4), 319-329. <https://doi.org/10.2111/REM-D-11-00141.1>
- Ganie, A.A., Mir, I.A., Sehgal, J.P., Sheikh, G.G., Haq, M.R., & Pampori, Z.A. (2011). Feeding systems-efficient tools to alter milk composition-A review. *Agricultural Reviews*, 32, 235-245.
- Garcia, L. D. S., Bettencourt, A. F., Aires, J. F., Andretta, I., & Fischer, V. (2023). The effect of diet restriction on raw milk stability: a meta-analytical approach. *The Journal of Dairy Research*, 90(4), 398-402. <https://doi.org/10.1017/S0022029923000705>
- Goyo, M. J. M., Gacutan Jr, M. D., & Galvez, L. A. (2025). A comparative study on the in vivo and in situ degradability of napier (*Pennisetum purpureum*), guinea (*Megathyrus maximus*), and paspalum (*Paspalum conjugatum*) as forage grasses. *International Journal of Multidisciplinary: Applied Business and Education Research*, 6(6), 2935-2942.
- Habimana, V., Nguluma, A. S., Nziku, Z. C., Charlotte, C., Morota, G., Mrode, R., & Chenyambuga, S. W. (2023). Heat stress effects on milk yield traits and metabolites and mitigation strategies for dairy cattle breeds reared in tropical and sub-tropical countries. *Frontiers in Veterinary Science*, 10, Article 1121499. <https://doi.org/10.3389/fvets.2023.1121499>
- Hayes, E., Wallace, D., O'Donnell, C., Greene, D., Hennessy, D., O'Shea, N., Tobin, J. T., & Fenelon, M. A. (2023). Trend analysis and prediction of seasonal changes in milk composition from a pasture-based dairy research herd. *Journal of Dairy Science*, 106(4), 2326-2337. <https://www.doi.org/10.3168/jds.2021-21483>
- Henno, M., Ots, M., Jõudu, I., Kaart, T., & Kärt, O. (2008). Factors affecting the freezing point stability of milk from individual cows. *International Dairy Journal*, 18(2), 210-215.
- Hernández-Castellano, L. E., Nally, J. E., Lindahl, J. F., Wanapat, M., Alhidary, I. A., Fanguero, D. P., Grace, D., Ratto, M. H., Bambou, J., & Almeida, A.M. (2019). Dairy science and health in the tropics: challenges and opportunities for the next decades. *Tropical Animal Health and Production*, 51, 1009-1017. <https://www.doi.org/10.1007/s11250-019-01866-6>
- Jaeger, A., Zannini, E., Sahin, A. W., & Arendt, E. K. (2021). Barley protein properties, extraction and applications, with a focus on brewers' spent grain protein. *Foods*, 10(6), Article 1389. <https://doi.org/10.3390/foods10061389>
- Johnson, C. E., Huber, J. T., & King, K. J. (1987). Storage and utilization of brewers wet grains in diets for lactating dairy cows. *Journal of Dairy Science*, 70, 98-107. [https://doi.org/10.3168/JDS.S0022-0302\(87\)79984-6](https://doi.org/10.3168/JDS.S0022-0302(87)79984-6)
- Kawakami, R., & Hasebe, H. (2019). Appropriate indoor environment for southeast asian people; physiological and psychological responses to humidity. *IOP Conference Series: Earth and Environmental Science*, 294(1), Article 012087. <https://doi.org/10.1088/1755-1315/294/1/012087>
- Kelly, A. L., Reid, S., Joyce, P., Meaney, W. J., & Foley, J. (1998). Effect of decreased milking frequency of cows in late lactation on milk somatic cell count, polymorphonuclear leucocyte numbers, composition and proteolytic activity. *The Journal of Dairy Research*, 65(3), 365-373. <https://doi.org/10.1017/s0022029998002921>
- Kondo, M., Yoshida, M., Loresco, M., Lapitan, R. M., Herrera, J. R. V., Barrio A. N. D., Uyeno, Y., Matsui, H., & Fujihara, T. (2015). Nutrient contents and in vitro ruminal

- fermentation of tropical grasses harvested in wet season in the Philippines. *Advances in Animal Veterinary Sciences*, 3(12), 694-699.
- Kumar, G., Devi, P., Sharma, N., & Somagond, Y. M. (2020). Impact of thermal stress on milk production, composition and fatty acid profile in dairy cows: A review. *Journal of Entomology and Zoology Studies*, 8, 1278-1283.
- Kung, L., Shaver, R. D., Grant, R. J., & Schmidt, R. J. (2018). Silage review: Interpretation of chemical, microbial, and organoleptic components of silages. *Journal of Dairy Science*, 101(5), 4020-4033. <https://doi.org/10.3168/jds.2017-13909>
- Lagrange, V., Whitsett, D. K., & Burris, C. (2015). Global market for dairy proteins. *Journal of Food Science*, 80, 6-22. <https://www.doi.org/10.1111/1750-3841.12801>
- Lalowski, P., Kruk, M., Salman, M., & Trzaskowska, M. (2024). Prebiotic properties of brewer's spent grain – literature review. *Food Science Technology Quality*, 3(140), 69-81. <https://doi.org/10.15193/zntj/2024/140/510>
- Li, H., Ran, Q., Jia, Z., Shuai, Y., Zhou, Q., & Guan, H. (2023). Effect of different dry matter content on fermentation characteristics and nutritional quality of Napier grass silage with novel lactic acid bacteria strains. *Letters in Applied Microbiology*, 76(2), Article ovad018. <https://doi.org/10.1093/lambio/ovad018>
- Li, X. (2020). Heat wave trends in Southeast Asia during 1979–2018: The impact of humidity. *Science of the Total Environment*, 721, Article 137664. <https://doi.org/10.1016/j.scitotenv.2020.137664>
- Linzell, J. L., & Peaker, M. (1971). The permeability of mammary ducts. *The Journal of Physiology*, 216(3), 701-716. <https://doi.org/10.1113/jphysiol.1971.sp009548>
- Lualhati, C. V., Turaja, K. I. B., Yu, L. C. P., Sandro, J. M., Maghirang, M. A. T., Espina, J. J. Jr., Nerona, C. M., Amido, R. D., Guia, M. R. B. U.-D., Guadayo, G. F., Loresco, M. M., & del Barrio, A. N. (2025). Influence of varying levels of *Lactiplantibacillus plantarum* inoculant on the physicochemical characteristics and degradation kinetics of corn silage. *Philippine Agricultural Scientist*, 108(3), 282-290. <https://doi.org/10.62550/HJ815060024>
- Mahnken, C. L. (2010). Utilization of wet brewers grains as a replacement for corn silage in lactating dairy cow diets. *Journal of Dairy Science*, 93, 871-871.
- Maxin, G., Ouellet, D., & Lapierre, H. (2013). Ruminal degradability of dry matter, crude protein, and amino acids in soybean meal, canola meal, corn, and wheat dried distillers grains. *Journal of Dairy Science*, 96(8), 5151-5160. <https://doi.org/10.3168/jds.2012-6392>
- McCabe, C. J., & Boerman, J. P. (2020). Invited review: Quantifying protein mobilization in dairy cows during the transition period. *Applied Animal Science*, 36(3), 389-396. <https://doi.org/10.15232/aas.2019-01929>
- McCarthy, O. J., & Singh, H. (2009). Physico-chemical properties of milk. In P. McSweeney, & P. F. Fox (Eds.). *Advanced Dairy Chemistry Vol 3 : Lactose, water, salts and minor constituents* (pp. 691-758). Springer Nature. https://doi.org/10.1007/978-0-387-84865-5_15
- Mitri, S., Salameh, J.-S., Khelfa, A., Leonard, E., Maroun, R. G., Louka, N., & Koubaa, M. (2022). Valorization of brewers' spent grains: Pretreatments and fermentation, a review. *Fermentation*, 8(2), Article 50. <https://doi.org/10.3390/fermentation8020050>
- Ndihokubwayo, F., & Koç, A. (2023). Breeding pure and crossbreeds of European cattle breeds for milk production improvement under tropical climate conditions-a review. *Tropical Animal Health and Production*, 56(1), Article 23. <https://doi.org/10.1007/s11250-023-03868-x>
- Nyameasem, J. K., Reinsch, T., Und, C. M., & Taube, F. (2018). *The potential of dairy production in sub-Saharan Africa*. https://www.lfl.bayern.de/mam/cms07/lipz/dateien/aggf_2018_nyameasem_et_al.pdf

- Nyhan, L., Sahin, A. W., Schmitz, H. H., Siegel, J. B., & Arendt, E. K. (2023). Brewers' spent grain: An unprecedented opportunity to develop sustainable plant-based nutrition ingredients addressing global malnutrition challenges. *Journal of Agricultural and Food Chemistry*, 71(28), 10543-10564. <https://doi.org/10.1021/acs.jafc.3c02489>
- Pacheco, D., Waghorn, G. C., & Janssen, P. H. (2014). Decreasing methane emissions from ruminants grazing forages: a fit with productive and financial realities? *Animal Production Science*, 54, 1141-1154. <https://doi.org/10.1071/AN14437>
- PCAARRD. (2024). *Dairy Industry Profile*. <https://ispweb.pcaarrd.dost.gov.ph/dairy/>
- Philippines Statistics Authority. (2023, August). *Dairy Situation Report, April to June 2023*. <https://psa.gov.ph/content/dairy-situation-report-april-june-2023>
- Polan, C. E., Herrington, T. A., Wark, W. A., & Armentano, L. E. (1985). Milk production response to diets supplemented with dried brewers grains, wet brewers grains, or soybean meal. *Journal of Dairy Science*, 68(8), 2016-2026. [https://doi.org/10.3168/jds.S0022-0302\(85\)81063-8](https://doi.org/10.3168/jds.S0022-0302(85)81063-8)
- Poli, S. (2004). The European community and the adoption of international food standards within the Codex Alimentarius Commission. *European Law Journal*, 10(5), 613-630.
- Promkot, C., & Wanapat, M. (2009). Effect of elemental sulfur supplementation on rumen environment parameters and utilization efficiency of fresh cassava foliage and cassava hay in dairy cattle. *Asian-Australasian Journal of Animal Sciences*, 22(10), 1366-1376.
- Reynolds, C. K., Aikman, P. C., Lupoli, B., Humphries, D. J., & Beever, D. E. (2003). Splanchnic metabolism of dairy cows during the transition from late gestation through early lactation. *Journal of Dairy Science*, 86(4), 1201-1217.
- Sammad, A., Wang, Y. J., Umer, S., Lirong, H., Khan, I., Khan, A., Ahmad, B., & Wang, Y. (2020). Nutritional physiology and biochemistry of dairy cattle under the influence of heat stress: Consequences and opportunities. *Animals*, 10(5), Article 793. <https://www.doi.org/10.3390/ani10050793>
- Samuelson, G., Bratteby, L.-E., Mohsen, R., & Vessby, B. (2001). Dietary fat intake in healthy adolescents: inverse relationships between the estimated intake of saturated fatty acids and serum cholesterol. *British Journal of Nutrition*, 85(3), 333-341. <https://doi.org/10.1079/BJN2000279>
- Santos, K. A., Stern, M. D., & Satter, L. D. (1984). Protein degradation in the rumen and amino acid absorption in the small intestine of lactating dairy cattle fed various protein sources. *Journal of Animal Science*, 58(1), 244-255. <https://doi.org/10.2527/jas1984.581244x>
- Sapru, A., Barbano, D. M., Yun, J. J., Klei, L. R., Oltenacu, P. A., & Bandler, D. K. (1997). Cheddar cheese: influence of milking frequency and stage of lactation on composition and yield. *Journal of Dairy Science*, 80(3), 437-446. [https://doi.org/10.3168/jds.s0022-0302\(97\)75955-1](https://doi.org/10.3168/jds.s0022-0302(97)75955-1)
- Sautier, M., Martin-Clouaire, R., Faivre, R., & Duru, M. (2013). Assessing climatic exposure of grassland-based livestock systems with seasonal-scale indicators. *Climatic Change*, 120, 341-355. <https://doi.org/10.1007/s10584-013-0808-2>
- Sauvant, D., Chapoutot, P., Peyraud, J.-L., Meschy, F., & Doreau, B. (2004). *Nutritional values for ruminants*. In D. Sauvant, J.-M. Perez, & G. Tran (Eds.). *Tables of composition and nutritional value of feed materials* (pp. 43-51). Wageningen Academic.
- Schingoethe, D. J. (1996). Balancing the amino acid needs of the dairy cow. *Animal Feed Science and Technology*, 60(3-4), 153-160.
- Shinozuka, Y., Kawai, K., Sato, R., Higashitani, A., Ueno, D., Okita, M., & Isobe, N. (2018). Effect of intramammary lipopolysaccharide infusion on milk pH of uninfused udder in goat. *Journal of Veterinary Medical Science*, 80(8), 1287-1290. <https://doi.org/10.1292/jvms.18-0280>
- Singh, A., Sidhu, S., & Singh, P. (2019). Bypass protein technology: A review. *The Pharma Innovation Journal*, 8(8), 150-153.

- Singh, V., & Qin, X. (2020). Study of rainfall variabilities in Southeast Asia using long-term gridded rainfall and its substantiation through global climate indices. *Journal of Hydrology*, 585, Article 124320. <https://doi.org/10.1016/j.jhydrol.2019.124320>
- Sloth, K. H., Friggens, N. C., Løvendahl, P., Andersen, P. H., Jensen, J., & Ingvarsen, K. L. (2003). Potential for improving description of bovine udder health status by combined analysis of milk parameters. *Journal of Dairy Science*, 86(4), 1221-1232. [https://doi.org/10.3168/jds.S0022-0302\(03\)73706-0](https://doi.org/10.3168/jds.S0022-0302(03)73706-0)
- Sørhaug, T., & Stepaniak, L. (1997). Psychrotrophs and their enzymes in milk and dairy products: Quality aspects. *Trends in Food Science and Technology*, 8(2), 35-41. [https://doi.org/10.1016/S0924-2244\(97\)01006-6](https://doi.org/10.1016/S0924-2244(97)01006-6)
- Sova, A., LeBlanc, S., McBride, B., & DeVries, T. (2013). Associations between herd-level feeding management practices, feed sorting, and milk production in freestall dairy farms. *Journal of Dairy Science*, 96(7), 4759-4770. <https://doi.org/10.3168/jds.2013-6679>
- Suadsong, S. (2012). Alleviating heat stress leads to improved cow reproductive performance. In N. Chaiyabutr (Ed.). *Milk production-an up-to-date overview of animal nutrition, management and health* (pp. 309-330). IntechOpen. <https://doi.org/10.5772/50881>
- Suhendra, D., Rahman, M. A., & Megawati, M. (2020). Correlation of protein contents and milk temperatures, with milk density of friesian holstein (fh) cow in ngablak district of magelang regency central java. *Journal of Livestock Science and Production*, 4(2), 289-294.
- Surendra, K. C., & Khanal, S. K. (2015). Effects of crop maturity and size reduction on digestibility and methane yield of dedicated energy crop. *Bioresource Technology*, 178, 187-193. <https://doi.org/10.1016/j.biortech.2014.09.055>
- Sutton, J. D., Dhanoa, M. S., Morant, S. V., France, J., Napper, D. J., & Schuller, E. (2003). Rates of production of acetate, propionate, and butyrate in the rumen of lactating dairy cows given normal and low-roughage diets. *Journal of Dairy Science*, 86(11), 3620-3633.
- Thanh, N. D., Thi, H. T., Quang, T. N., Van, H. N., Nguyen, G. H., Huu, Q. N., & Son, T. T. (2023). Performance evaluation of multiple particulate matter monitoring instruments under higher temperatures and relative humidity in Southeast Asia and design of an affordable monitoring instrument (ManPMS). *Instrumentation Science and Technology*, 51(6), 660-680. <https://doi.org/10.1080/10739149.2023.2193635>
- Thirumalai, K., DiNezio, P. N., Okumura, Y., & Deser, C. (2017). Extreme temperatures in Southeast Asia caused by El Niño and worsened by global warming. *Nature Communications*, 8(1), Article 15531. <https://doi.org/10.1038/ncomms15531>
- Tomé, D., Benoit, S., & Azzout-Marniche, D. (2020). Protein metabolism and related body function: mechanistic approaches and health consequences. *Proceedings of the Nutrition Society*, 80, 243-251. <https://doi.org/10.1017/S0029665120007880>
- Tsioulpas, A., Grandison, A. S., & Lewis, M. J. (2007a). Changes in physical properties of bovine milk from the colostrum period to early lactation. *Journal of Dairy Science*, 90(11), 5012-5017. <https://doi.org/10.3168/jds.2007-0192>
- Tsioulpas, A., Lewis, M. J., & Grandison, A. S. (2007b). A study of the pH of individual milk samples. *International Journal of Dairy Technology*, 60, 96-97. <https://doi.org/10.1111/J.1471-0307.2007.00308.X>
- Urrutia, N. L., & Harvatine, K. J. (2017). Acetate dose-dependently stimulates milk fat synthesis in lactating dairy cows. *The Journal of Nutrition*, 147(5), 763-769. <https://doi.org/10.3945/jn.116.245001>
- Vaiglova, P., Ávila, J. N., Buckley, H., Galipaud, J. C., Green, D. R., Halcrow, S., James, H. F., Kinaston, R., Oxenham, M., Paz, V., Simanjuntak, T., Snoeck, C., Trinh, H. H.,

-
- Williams, I. S., & Smith, T. M. (2024). Past rainfall patterns in Southeast Asia revealed by microanalysis of $\delta^{18}\text{O}$ values in human teeth. *Journal of Archaeological Science*, 162, Article 105922. <https://doi.org/10.1016/j.jas.2023.105922>
- Vera, J. C., Ruiz, A., Mayo, R., Ayala, A., Aguilar, C., & Solorio, F. J. (2014). Manipulation of the energy metabolism of ruminants in the tropics: Options for improving meat and milk production and quality. *Cuban Journal of Agricultural Science*, 48, 43-53.
- Walstra, P., Wouters, J. T. M., & Geurts, T. J. (2006). *Dairy Science and Technology*, (2nd Ed.). Taylor and Francis Group, Boca Raton, London
- West, J. W., Ely, L. O., & Martin, S. A. (1994). Wet brewers grains for lactating dairy cows during hot, humid weather. *Journal of Dairy Science*, 77(1), 196-204. [https://doi.org/10.3168/jds.S0022-0302\(94\)76942-3](https://doi.org/10.3168/jds.S0022-0302(94)76942-3)
- Westendorf, M. L., & Wohlt, J. E. (2002). Brewing by-products: their use as animal feeds. *The Veterinary Clinics of North America. Food Animal Practice*, 18(2), 233-252. [https://doi.org/10.1016/s0749-0720\(02\)00016-6](https://doi.org/10.1016/s0749-0720(02)00016-6)
- Wheelock, J., Rhoads, R., VanBaale, M., Sanders, S., & Baumgard, L. (2010). Effects of heat stress on energetic metabolism in lactating Holstein cows. *Journal of Dairy Science*, 93(2), 644-655. <https://doi.org/10.3168/jds.2009-2295>
- White, J. S., Yohannan, B. K., & Walker, G. M. (2008). Bioconversion of brewer's spent grains to bioethanol. *FEMS Yeast Research*, 8(7), 1175-1184. <https://doi.org/10.1111/j.1567-1364.2008.00390.x>
- Ysulat, R. I., Migalbin, J. R., & Torres, R. O. (2024). Impact of fertilization on productivity and nutritional quality of napier varieties in Southern Mindanao, Philippines. *International Journal of Plant and Soil Science*, 36(10), 318-328.